

Examiner reports

Q2.

Part (a) was the most accessible part of this question. While there were a good number of correct responses, there were also a significant number of students who had problems with the initial position. Often this was initially placed on the wrong side of the equals sign so that the students ended up with $-(500\mathbf{i} + 200)$ in their expressions for the position vector.

Part (b) was much more challenging for the students. Those who were able to form two quadratic equations from the components of the position vector were usually able to go on and solve these to find the correct time. Incorrect approaches included trying to work with the magnitude of the position vector or not using the position vector of the rock.

Part (c) was also problematic for some students. Many ended up with $\frac{300\mathbf{i} + 600\mathbf{j}}{\text{time}}$ instead of $\frac{-300\mathbf{i} - 600\mathbf{j}}{\text{time}}$.

The explanations in part (d) were generally inadequate, with answers like velocity has direction and speed does not, which did not go far enough to justify the award of the mark.

Q3.

There were some very varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).

Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement method led candidates to the correct answer.

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Q5.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west. Part (c) was found to be more demanding. Candidates who did not identify the $\mathbf{i}$ and $\mathbf{j}$ components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions. Some forgot to square the speed and tried to work from equations such as $(4 - 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be $-3\mathbf{i} + 4\mathbf{j}$, which produced a speed of 5.

Q6.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the $+\mathbf{c}$ term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding $m\mathbf{a}$, which was $2(6t\mathbf{i} - 8\mathbf{j})$, to be $12t\mathbf{i} - 8\mathbf{j}$.

Q7.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like

$500 = \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j})^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q8.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of $\mathbf{F}$, giving instead just the vector $\mathbf{F}$, $-40\mathbf{i} + 30\mathbf{j}$. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the $+\mathbf{c}$ term and thus to ignore the initial position vector. Some found the $+\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was the initial position vector.

Q9.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example

working such as: $m = \frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6\mathbf{i} + 6\mathbf{j}$. Candidates should be advised to consider the

components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector $3\mathbf{i} + 4\mathbf{j}$ as their final answer and did not find the magnitude in order to get the distance as requested.

Q10.

Most of this question was answered well. In part (a), the only difficulty was found in

differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or $-3.52\mathbf{i}$], but did not find the speed of $+3.52 \text{ m s}^{-1}$. Others gave the speed as 3.5 m s^{-1} , which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q11.

This question was demanding for the candidates, although part (a) caused very few difficulties for the majority, and some gained only the three marks for this part of the question.

Part (b) was the most demanding part of the question, and only very few candidates gained the marks available. Many candidates stated that the $\mathbf{i}$ component should be zero, but very few formed an equation to find the time when this would be. Most incorrect approaches were based on using a time, often 20 seconds, and then calculating the $\mathbf{j}$ component of the velocity and stating that the $\mathbf{i}$ component was 0.

There were some good answers to part (c), but quite a few candidates failed to include the initial position of the particle.

In part (d), there were some good solutions, but some candidates made errors. These

included not subtracting the initial position. Some candidates used $\frac{\mathbf{u} + \mathbf{v}}{2}$. With this method there were two main errors: dividing by 20 rather than 2 and using an incorrect expression for either $\mathbf{u}$ or $\mathbf{v}$.

Q12.

In this question the use of vector notation by some candidates was quite poor. Part (a) was done well by a good number of candidates, but some did not show enough working to gain full marks. Typical issues were not showing that their equation was derived from $\mathbf{v} = \mathbf{u} + \mathbf{at}$ and not showing the division by 40 clearly. Some candidates did well with part (b), but there were quite a lot of arithmetic errors.

Some candidates used an incorrect initial velocity, for example $4\mathbf{i}$ instead of $5\mathbf{j}$. Part (c) proved to be too demanding for the majority of the candidates, although some were able to gain a few marks. One of the most common errors was to work with position vectors rather than with velocities. For those who did work with velocities, creating an equation to find the time when the Jet Ski was travelling south east was found to be difficult. Quite a number of the candidates who got this far had equations that lacked a minus sign.

Q13.

This question was generally answered well. The only common error was in part (b)(ii), where a number of candidates found $\mathbf{F}$ to be $36\mathbf{i} + 6\mathbf{j}$, but did not find its magnitude.

Q14.

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.

Q15.

Many candidates made good attempts at parts (a) and (b). The majority of errors were due to errors made manipulating the vectors. Some candidates did write down correct equations or expressions to begin their solutions, but made errors in manipulating their expressions. In part (b), a few candidates included a $75\mathbf{i}$ term.

Several candidates simply did not attempt part (c) of the question. Of those that did, some tried to find a time using expressions for the position of the particle rather than its velocity, while other candidates tried to use the equation $v^2 = u^2 + 2as$.

Q16.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q17.

Most candidates completed this question successfully. A few used $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ in part (b) and obtained $\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 6t^2)\mathbf{j}$, rather than the given answer. After they had done the integration of $\mathbf{a}$, a number of candidates 'invented' the $6\mathbf{i} + 30\mathbf{j}$ terms required.

Q18.

This was found to be the most difficult question on the paper. While the most able were able to score full marks, others showed confusion between vectors and scalars, or simply

showed misunderstanding of vector quantities. Many found the vector $\vec{AB}$ in part (a) but could not then proceed correctly, with ratios of vectors often leading to vector expressions for the time. The most successful methods included the application of the equations of motion with constant acceleration, and the use of diagrams with explained steps in motion. Part (b) often began with the assumption of a zero initial velocity, some only considered the $\mathbf{j}$ component of the motion, and many failed to add the position vector of the particle at B. Part (c) often began well with the vector $\vec{AC}$ found, but with no attempt to find its magnitude to give the distance requested.

Q19.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting $t = 1/3$ correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that $\mathbf{j}$ was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of $t = 4$.

Q20.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

Q21.

Part (a) was often done well, but a number of candidates effectively repeated the same assumption twice. The candidates were often helped by the printed answer in part (b). Many of the candidates made life difficult for themselves by finding the initial speed and angle of projection and then working with these values. One consequence of this was that the final answers were inaccurate. Part (c) proved more challenging for candidates. The most common incorrect approach was to assume that the ball hit the wall when it was at its maximum height. In part (d) there were some good answers and many of the candidates were on the right track. However quite a few of the candidates produced answers which lacked clarity of explanation.

Q22.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q23.

Candidates produced good work in this question, and were able to score marks on the parts where they had the necessary knowledge and understanding. Part (a)(i) was done well, and part (a)(ii), although an unfamiliar request, produced many good solutions. Part (a)(iii) was done well in a variety of ways. Parts (b) and (c) proved successful for most candidates.

Q24.

Most could attempt differentiation correctly to find an expression for the velocity in part (a).

However poor algebra and lack of appreciation of the form of scalar quantities let down many in part (b). Correct answers in part (b) were often then reduced to ' $6 + 2t$ ' in part (c) and few completed the request correctly.

Q25.

It should be noted that vector notation is essential throughout such questions as this, whether in the form of the unit vectors i and j , or in column vector form. Part (a) was usually started well, but too often the constant of integration was missing. Part (c) tested understanding and proved discriminating as expected.

Q26.

Parts (a) and (b) were done well by the better prepared candidates, but few spotted and used the pattern of decreasing distances in part (c).

Q27.

The technique of integration was correctly selected and performed accurately by most candidates. Many then completed part (a) correctly, but a minority omitted the constant of integration and failed to use the initial conditions to establish the correct position vector. Part (b) was often fully correct, although some candidates added the position vectors of C and D rather than subtracting them in their attempt to find the vector $\overrightarrow{CD}$.

Q28.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most

popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at O or at C , the request was straightforward and full marks were often obtained.

Q29.

Part (a) showed an improvement in handling vectors in this form and also in differentiation techniques. Part (b) proved discriminating, as intended, and many failed to correctly link the direction of motion with the velocity components. Many candidates were successful in finding a correct expression for the acceleration in part (c), but not all were then able to correctly interpret their expression.

Q30.

Part (a) of this question was generally done very well, although some unusual vector manipulation was observed on a few scripts. Several candidates worked with each component and combined their results into a vector, rather than working with vectors throughout their solutions.

In part (b), a large number of candidates found the velocity correctly, but did not then go on to find the speed of the particle.

There were a fair number of good attempts at part (c). Some candidates made errors with the signs when first setting up their equations. There were a few candidates who did not make any significant attempt at this part of the question.

Q31.

Part (a) was generally answered well with a mixture of integration and constant acceleration equation methods.

Part (b) was done better than expected, with only a small number of candidates working with the wrong component.

There were a variety of minor errors in part (c), but also many correct solutions.

Many candidates found part (d) too demanding and made little progress. It was not uncommon to see solutions that still contained an acceleration or that ignored the initial position of the particle.

Q32.

Many candidates were able to complete part (a), probably helped by the printed answer. Many of these candidates did not produce any further correct working. In part (b), some candidates could write down correct expressions for the position vector. A common error was to apply the constant acceleration formula $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$ using the two velocities given in the question without realising that $\mathbf{v}$ was variable.

Of the candidates who had a correct answer in part (b), a fair proportion went on to try part (c). There were a reasonable number of correct responses, but there were a number who used the wrong component or a mixture of the two. There were very few correct responses to part (d). The main incorrect approach was to use the position vector. Those

candidates who realised they needed to find an expression for the velocity generally went on to complete this part of the question.

Q33.

Parts (a) and (b) proved to be quite accessible for most candidates, but part (c) was very much more demanding. In part (a) the candidates tended to score all or no marks, with a complete solution following a correct first step. There were occasional arithmetic errors in the working of some candidates. A correct answer to part (a) was often followed by the velocity in part (b). A few candidates used incorrect formulae or tried to use an expression for the position vector rather than the velocity. Part (c) caused a variety of problems, with some candidates forming equations that contained vectors. A few produced an equation that would have arisen from moving south west instead of south east. A correct equation was normally followed by a complete solution. A very small number of candidates found the velocity, but not the speed of the boat.

Q34.

This question was done well by quite a few candidates. Many of these found the acceleration as part of their solution to part (a). The candidates who had difficulties were those who did not realise that they needed to use a constant acceleration formula or those who quoted an inappropriate formula as their starting point. A few candidates did have difficulty with part (b), but several who had struggled with part (a) used the printed result to enable them to attempt part (b). Those who had been successful with part (b) normally went on to complete part (c).

Q35.

Part (a) was done very well by a good number of the candidates. A few worked with the magnitudes of the quantities, starting with the magnitude of the force, rather than with the actual vectors. Part (b) was found to be much more demanding. The two main errors made by the candidates were to assume that the particle still had the same acceleration as in part (a) or to use a time of 45 s rather than 25 s for the second stage of the motion.

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